

CODE SPORT



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In this month's column, we continue our discussion on content moderation using NLP techniques, focusing on hate speech detection in social media textual data.

Last month, we had discussed how a neural network classifier based on convolutional neural network architecture can be used to detect offensive content in social media posts. Let's continue the discussion, looking at some of the practical challenges associated with detecting offensive content in social media posts.

There have been a number of efforts in developing automated machine learning (ML) and natural language processing (NLP) based techniques for detecting hate speech and offensive content in social media. Given that much of human conversational language is implicit, textual data carries considerable assumptions of common sense and contextual world knowledge. Hence, with automated approaches, it may be difficult to disambiguate between offensive content and non-offensive content. For example, consider the sentence, "Get a life!" While this sentence contains no profanity, it is still considered offensive by humans. Similarly, an automated approach may find it difficult to disambiguate between "Your hand is dirty" and "You have a dirty mouth!" The first sentence is likely to be non-offensive as it describes an actual situation, whereas the second sentence is more likely to be an implicit toxic comment indicating that the target indulges in bad language. Given the similarity of the two sentences, most automated approaches either label both as offensive content or mark both as non-offensive. The implicit common sense knowledge that 'dirty mouth' can be an allusion to someone speaking foul language is not available to the automated approaches, which results in misclassification.

Given the widely varying vocabulary of human conversations, it is not possible for purely automated approaches to catch all cases of offensive content in social media. Therefore, a more practical approach is to design a hybrid system consisting of both automated techniques and 'humans in the loop'. The intent of the automated techniques should be to aid the humans in

the loop by reducing the content moderation workload, such that they can filter out instances that they can confidently predict the label for, while passing on the ambiguous cases to the human moderators. The automated approaches can also aid the human moderators by providing additional information to help them classify the content by performing the following tasks:

- Highlighting the key words or passages that the automated approach may predict as abusive content.
- Linking external information about entities, events, terms mentioned in the textual content, etc, about which the human moderators may be unaware of.

Recall that human moderation is an expensive and laborious task. Hence, many social media companies typically outsource this task to back-end operations situated in far-flung countries. For instance, social media companies in USA may employ a team of human moderators sitting in the Philippines or in India. Given the socio-cultural differences between the people in India and Philippines, it would be difficult for the human moderators to flag ambiguous cases that may make use of local 'lingo'. For example, consider the sentence, "You are daft as a bush." This has no explicit profanity and each individual word by itself is not offensive. However, the phrase is typically used in British English as slang for someone who is crazy or confused. Similarly, consider the sentence, "Have you seen his house? It has gone to the dogs." This again is an offensive comment containing an idiomatic phrase which a human moderator from another social milieu may not always be aware of.

Apart from linguistic difficulties, there are also issues due to non-familiar entities and events being referenced in the input text, which the moderator may have very little knowledge of. Consider the example, "You act like a lemming!" The reference to a lemming may not be obvious to all human moderators since it